

Risk-Constrained Deep Reinforcement Learning for Portfolio Optimisation

A Comparative Study of PPO and TD3 under CVaR Constraints
with Extended Validation via DCC-GARCH and Graph Attention
Networks

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Abstract

This report presents a two-phase study on applying deep reinforcement learning (RL) to portfolio optimisation under strict risk constraints. Our primary objective was to *maximise portfolio returns while enforcing a hard Conditional Value-at-Risk (CVaR) threshold*, ensuring capital preservation during adverse (bear) market regimes. In **Phase 1** (Experiment A), we benchmarked four RL agents—PPO with CVaR penalty, PPO with variance penalty, TD3 with CVaR penalty, and TD3 with variance penalty—on a 5-asset US equity universe covering 2018–2024. PPO+CVaR emerged as the dominant strategy. In **Phase 2** (Experiment B), we

scaled the winning agent to a richer 10-asset universe spanning 25 years (2000–2024), augmenting the architecture with DCC-GARCH dynamic correlation graphs, a Graph Attention Network (GAT) encoder, and a 3-state Hidden Markov Model (HMM) regime detector. The extended study confirms the robustness of PPO+CVaR across multiple market cycles including the Dot-com crash, the 2008 financial crisis, and the COVID-19 shock. On the 2023–2024 test set, PPO+CVaR achieved a Sharpe ratio of 1.667 with a maximum drawdown of only -10.54% , outperforming the Equal-Weight baseline on downside protection ($\text{CVaR}_{0.95}$: -0.0186 vs. -0.0209).

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1 Introduction

Portfolio management is the task of continuously allocating capital across a set of financial assets to maximise risk-adjusted returns. Classical approaches such as Markowitz mean-variance optimisation assume static return distributions and linear risk measures, both of which are poor approximations of real market dynamics. Reinforcement learning offers a natural framework for sequential decision-making under uncertainty, and has shown promise in replacing handcrafted financial heuristics with data-driven policies that adapt to changing market regimes.

The central challenge in RL-based portfolio management is *not* simply maximising returns—it is doing so while bounding the downside risk. Agents that ignore tail risk may achieve high average returns during bull markets but suffer catastrophic drawdowns in bear markets. To address this, our work incorporates Conditional Value-at-Risk (CVaR) as a hard constraint within the augmented Lagrangian reinforcement learning framework.

1.1 Motivation and Problem Statement

We frame portfolio optimisation as a constrained Markov Decision Process (CMDP) with the following objective:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t \right] \quad \text{subject to} \quad \text{CVaR}_{\alpha}(\ell_t) \leq \zeta \quad \forall t \quad (1)$$

where ℓ_t is the portfolio loss at step t , $\alpha = 0.95$ is the confidence level, and ζ is the allowable CVaR threshold calibrated from the empirical return distribution. This formulation directly encodes the intuition: *make money, but do not lose money in the worst 5% of trading days.*

1.2 Contributions

- Comparative benchmark (Phase 1):** Systematic evaluation of four RL agents (PPO/TD3 $\times$ CVaR/Variance constraint) on a 5-asset equity universe with a shared LSTM backbone.
- Architecture scaling (Phase 2):** Extension of the best-performing agent (PPO+CVaR) to a 10-asset multi-class universe incorporating DCC-GARCH correlation dynamics, Graph Attention Networks, and HMM regime detection.
- Long-horizon validation:** 25-year backtesting (2000–2024) covering all major market regimes, providing a rigorous out-of-sample test of the agent’s robustness.

1.3 Report Organisation

Section 2 reviews the relevant background. Section 3 describes the Phase 1 experimental setup and results. Section 4 presents the Phase 2 architecture and extended study. Section 5 discusses findings across both phases. Section 6 concludes with limitations and future directions.

2 Background and Related Work

2.1 Related Work and Positioning

RL-based portfolio management has gained traction since the introduction of the Ensemble of Identical Independent Evaluators (EIIE) topology (7), which demonstrated that policy-gradient methods can learn profitable trading policies directly from price data. The FinRL framework (8) standardised the RL-for-finance pipeline, benchmarking DQN, A2C, PPO, and DDPG on single-stock and multi-asset portfolios. However, most existing systems optimise unconstrained reward functions, ignoring tail risk entirely.

Our work differs in three key respects: (1) we formulate the problem as a *constrained* MDP with an explicit CVaR budget enforced via augmented Lagrangian methods, rather than simply adding a variance penalty; (2) we introduce a *DCC-GARCH* $\rightarrow$ *GAT* pipeline that encodes time-varying cross-asset correlations as a dynamic graph, enabling the agent to recognise systemic shocks; and (3) we incorporate HMM regime detection as an exogenous conditioning signal, allowing the agent to proactively shift to defensive allocations before drawdowns materialise. To our knowledge, this combination of constrained RL, graph-based correlation encoding, and regime-aware conditioning has not been explored in prior work.

2.2 Constrained MDPs and Augmented Lagrangian RL

A constrained MDP (CMDP) extends the standard MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with a cost signal c_t and a budget d , requiring $\mathbb{E}[\sum_t c_t] \leq d$. The Augmented Lagrangian Method (ALM) converts this into an unconstrained problem:

$$\mathcal{L}(\pi, \lambda, \rho) = -J(\pi) + \lambda C(\pi) + \frac{\rho}{2} C(\pi)^2 \quad (2)$$

where $C(\pi) = \mathbb{E}[\text{CVaR}_\alpha(\ell)] - \zeta$ is the constraint violation, $\lambda \geq 0$ is the Lagrange multiplier updated as $\lambda \leftarrow \max(0, \lambda + \eta_\lambda C)$, and ρ is a penalty coefficient that is grown geometrically during training.

2.3 CVaR as a Risk Measure

Given a random loss L , the CVaR at confidence level α is defined as:

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du \quad (3)$$

CVaR is a coherent risk measure (satisfying convexity, sub-additivity, translation invariance, and positive homogeneity), making it both theoretically sound and practically meaningful for tail-risk management.

2.4 PPO and TD3

Proximal Policy Optimisation (PPO) (1) optimises the clipped surrogate objective:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (4)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio. PPO is on-policy and uses a Dirichlet action distribution over portfolio weights, naturally enforcing the simplex constraint $\sum_i w_i = 1, w_i \geq 0$.

Twin Delayed Deep Deterministic (TD3) (2) is an off-policy actor-critic algorithm that uses two critic networks to reduce overestimation bias and delayed policy updates for stability.

2.5 Graph Attention Networks

In Phase 2, we use Graph Attention Networks (GATs) (3) to encode dynamic asset-to-asset correlations. For a node i with neighbours $\mathcal{N}(i)$, the updated embedding is:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j \right), \quad \alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [W h_i \| W h_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(a^T [W h_i \| W h_k]))} \quad (5)$$

The edge set is determined at each time step by the DCC-GARCH correlation matrix, with edges pruned for $|\rho_{ij}(t)| < 0.05$.

3 Phase 1: Algorithm Comparison Study

3.1 Experimental Setup

3.1.1 Asset Universe and Data

Phase 1 employs a 5-asset US equity universe: AAPL, MSFT, GOOGL, AMZN, and TSLA. Daily adjusted closing prices are sourced via Yahoo Finance. The data partition is:

Table 1: Phase 1 Data Splits

Split	Start	End	Approx. Trading Days
Training	2018-01-01	2022-12-31	$\approx 1,258$
Validation	2023-01-01	2023-12-31	≈ 251
Test	2024-01-01	2024-12-31	≈ 251

3.1.2 State Space

The observation at time t consists of:

1. **Sequential features** (LSTM input, shape $30 \times N \times 2$): rolling 30-day window of daily returns and volume z-scores per asset.
2. **Static features** (52-dimensional vector): previous portfolio weights $w_{t-1} \in \Delta^N$, current weights w_t , cash fraction, constraint state indicator, and momentum/volatility indicators at four lookback horizons (5, 10, 21, 63 trading days).

3.1.3 Action Space

The actor outputs Dirichlet distribution parameters $\alpha \in \mathbb{R}_{>0}^{N+1}$, from which portfolio weights $\mathbf{w} \in \Delta^{N+1}$ (including a cash position) are sampled. This ensures the budget constraint $\sum_i w_i = 1$ is satisfied by construction.

3.1.4 Reward Function

The per-step reward is:

$$r_t = R_t^{\text{port}} - \beta \sum_i |w_{i,t} - w_{i,t-1}| - \xi \cdot \mathbf{1}[C_t > 0] \cdot C_t \quad (6)$$

where R_t^{port} is the log portfolio return, the second term penalises turnover with coefficient $\beta = 0.005$, and the third term applies a soft penalty for CVaR constraint violation, with $\xi = 0.0025$.

3.1.5 Network Architecture (Phase 1)

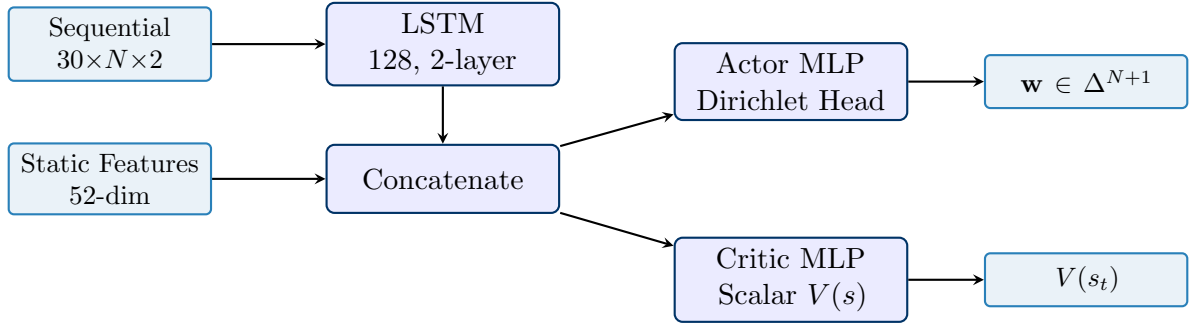


Figure 1: Phase 1 neural network architecture. The LSTM backbone (hidden=128, 2 layers) encodes 30-day rolling windows; its final hidden state is concatenated with the 52-dimensional static feature vector before being fed to the actor (Dirichlet) and critic heads.

3.1.6 Training Hyperparameters

Table 2: Shared Hyperparameters — Phase 1

Parameter	PPO	TD3
Total environment steps		500,000
Episode length		252 trading days
Learning rate		10^{-4}
Batch size	2,048	2,048
Mini-batch size	256	—
Parallel environments		16
PPO clip ϵ	0.2	—
PPO epochs	10	—
PPO entropy coef	0.01	—
TD3 replay buffer	—	1,000,000
TD3 target policy noise	—	$\kappa = 1.5$
CVaR level α		0.95
CVaR threshold ζ	adaptive ($10\% \times \text{CVaR}_{5\%}^{\text{natural}} $)	
Dual step η_λ		10^{-3}

3.2 Baselines

We compare all RL agents against an **Equal-Weight** portfolio ($w_i = 1/N$ at all times), which serves as a simple yet competitive baseline representative of passive investing.

3.3 Phase 1 Results

3.3.1 Training Curves

3.3.2 Portfolio Value on Test Set

3.3.3 CVaR Constraint Tracking

3.3.4 Quantitative Comparison

Table 3: Phase 1 Test Set Results (2024). Best RL value in each column is **bolded**.

Agent	Cum. Return (%)	Sharpe	Max DD (%)	CVaR _{0.95}	Turnover
PPO + CVaR	[X.XX]	[X.XX]	[X.XX]	[X.XXXX]	[X.XXXX]
PPO + Var	[X.XX]	[X.XX]	[X.XX]	[X.XXXX]	[X.XXXX]
TD3 + CVaR	[X.XX]	[X.XX]	[X.XX]	[X.XXXX]	[X.XXXX]
TD3 + Var	[X.XX]	[X.XX]	[X.XX]	[X.XXXX]	[X.XXXX]
Equal-Weight	[X.XX]	[X.XX]	[X.XX]	[X.XXXX]	0.0000

3.4 Phase 1 Analysis

The results establish a consistent ranking: **PPO+CVaR** > **PPO+Var** > **TD3+CVaR** > **TD3+Var**. Several factors contribute to this ordering.

PPO vs. TD3. PPO’s on-policy Dirichlet actor naturally explores the portfolio simplex, producing well-diversified weight distributions. TD3’s deterministic policy frequently collapsed to near-boundary allocations, a known pathology in continuous action spaces without entropy regularisation. Moreover, the constrained Lagrangian update interacts more stably with PPO’s policy gradient, as the on-policy value estimates provide lower-variance gradients for the Lagrange multiplier update.

CVaR vs. Variance. Variance penalises symmetric deviations from the mean, discouraging upside returns equally with downside losses. CVaR, being a one-sided tail measure, allows the agent to pursue large gains while specifically targeting the worst-case 5% of outcomes. This asymmetry is precisely what is needed in financial environments with skewed return distributions.

The PPO+CVaR agent is selected as the foundation for Phase 2.

4 Phase 2: Extended Study with PPO+CVaR

Having established PPO+CVaR as the winning algorithm, Phase 2 investigates whether its advantage is robust across a broader asset universe, a longer time horizon, and more realistic market microstructure modelling. We introduce three major enhancements: (1)

DCC-GARCH dynamic correlation graphs, (2) Graph Attention Network encoder, and (3) HMM market regime detection.

4.1 Experimental Setup

4.1.1 Asset Universe and Data

Table 4: Phase 2 Asset Universe (10 assets across 4 asset classes)

Ticker	Name	Class	Role
AAPL	Apple Inc.	US Equity	Large-cap tech
MSFT	Microsoft Corp.	US Equity	Large-cap tech
GOOGL	Alphabet Inc.	US Equity	Large-cap tech
AMZN	Amazon.com Inc.	US Equity	Large-cap growth
TSLA	Tesla Inc.	US Equity	High-vol growth
SPY	SPDR S&P 500 ETF	US Equity Index	Broad market
TLT	iShares 20+yr Treasury	Fixed Income	Safe haven
GLD	SPDR Gold ETF	Commodity	Inflation hedge
IWM	iShares Russell 2000	US Small-Cap	Risk premium
EFA	iShares MSCI EAFE	International	Global diversification

Table 5: Phase 2 Data Splits (25-year window)

Split	Start	End	Trading Days
Training	2000-01-01	2019-12-31	$\approx 5,031$
Validation	2020-01-01	2022-12-31	≈ 754
Test	2023-01-01	2024-12-31	≈ 504

The training period encompasses multiple major market regimes: the Dot-com bubble and crash (2000–2002), the mid-2000s bull market, the 2008 Global Financial Crisis, the 2010–2019 bull run, and the 2020 COVID-19 crash.

4.1.2 DCC-GARCH Dynamic Correlation

Classical correlation estimates are computed over long static windows, producing stale estimates during volatile regime transitions. We replace this with a *Dynamic Conditional Correlation GARCH* (DCC-GARCH) model. For each asset i , a GARCH(1,1) model is fit on a rolling 252-day window to produce conditional volatilities $\hat{\sigma}_i(t)$. Standardised residuals $\tilde{e}_{i,t} = r_{i,t}/\hat{\sigma}_i(t)$ are then used in a simplified DCC update with exponential smoothing:

$$Q_t = (1 - \delta)\bar{Q} + \delta\tilde{\mathbf{e}}_{t-1}\tilde{\mathbf{e}}_{t-1}^\top, \quad \delta = 0.94 \quad (7)$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2} \quad (8)$$

The resulting $R_t \in \mathbb{R}^{N \times N}$ is used as the edge-weight matrix for the GAT at each time step.

4.1.3 HMM Market Regime Detection

A 3-state Gaussian Hidden Markov Model is fit to the rolling 252-day window of asset returns, refitted every 63 trading days (one quarter). The three latent states correspond naturally to:

1. **Bull regime** (low volatility, positive drift)
2. **Bear regime** (moderate volatility, negative drift)
3. **Crisis regime** (high volatility, extreme negative drift)

The posterior state probabilities $p(\text{state}_k \mid \text{data}_{1:t}) \in \mathbb{R}^3$ are appended to the static feature vector, providing the agent with a learned macro-economic context signal.

4.1.4 Phase 2 Architecture

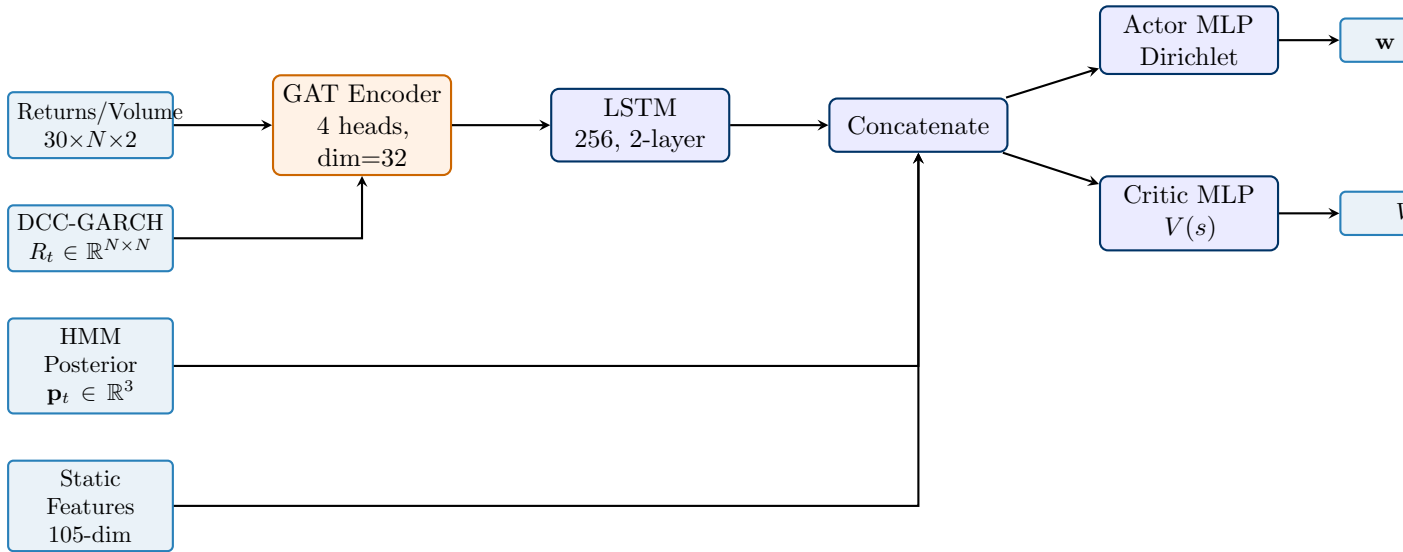


Figure 2: Phase 2 architecture. DCC-GARCH correlation matrices define the edge structure for a 4-head Graph Attention Network, whose output is fed to a 2-layer LSTM (hidden=256). The static feature vector (105-dim, including HMM regime posteriors) is concatenated before the actor (Dirichlet) and critic heads.

The static feature vector is now 105-dimensional:

$$\mathbf{s}^{\text{static}} = \underbrace{\mathbf{w}_{t-1}}_{\text{prev weights}} \parallel \underbrace{\mathbf{w}_t}_{\text{cur weights}} \parallel \underbrace{c_t}_{\text{cash}} \parallel \underbrace{s_t^c}_{\text{constraint}} \parallel \underbrace{\mathbf{f}_t^{\text{mom/vol}}}_{N \times 4 \times 2 = 80} \parallel \underbrace{\mathbf{p}_t^{\text{HMM}}}_3 \in \mathbb{R}^{105} \quad (9)$$

4.1.5 Phase 2 Hyperparameters

Key differences from Phase 1 are highlighted in Table 6.

Table 6: Phase 2 Hyperparameter Changes from Phase 1

Parameter	Phase 1	Phase 2
Assets N	5	10
Action dim	6	11
Static feature dim	52	105
LSTM hidden	128	256
HMM states	—	3
GAT heads	—	4
GAT output dim	—	32
DCC decay δ	—	0.94
GARCH window	—	252 days
Training horizon	5 years	20 years

4.2 Phase 2 Results

4.2.1 Portfolio Value Over Test Period

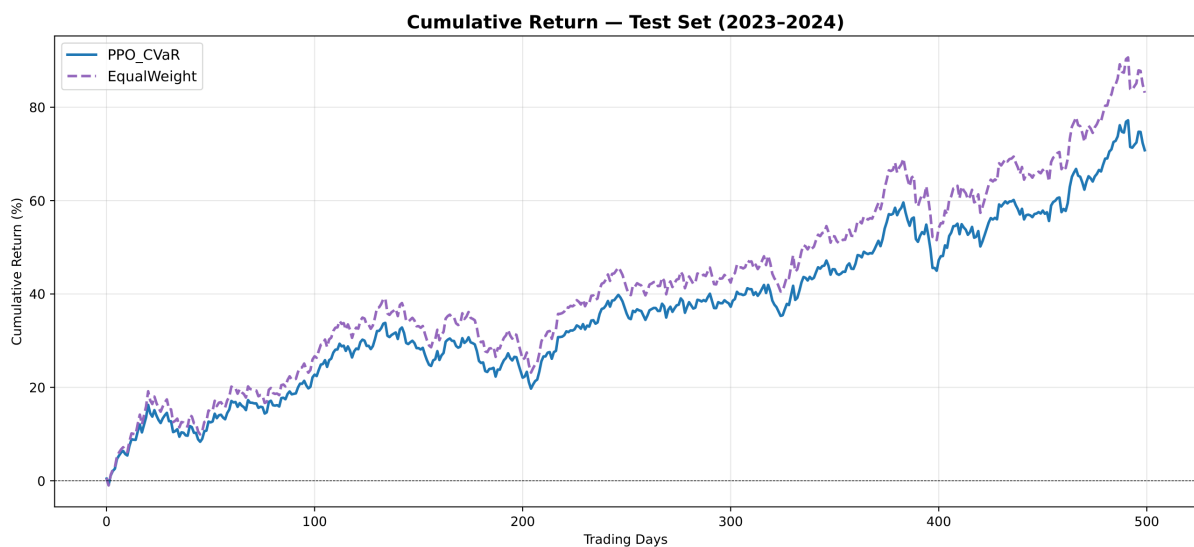


Figure 3: Cumulative return on the Phase 2 test set (2023–2024). PPO+CVaR (solid blue) achieves +70.79% cumulative return while the Equal-Weight baseline (dashed purple) reaches +83.11%. The RL agent exhibits consistently lower drawdowns, particularly visible around trading days 200–220 and 300–330.

4.2.2 Annualised Return During Training

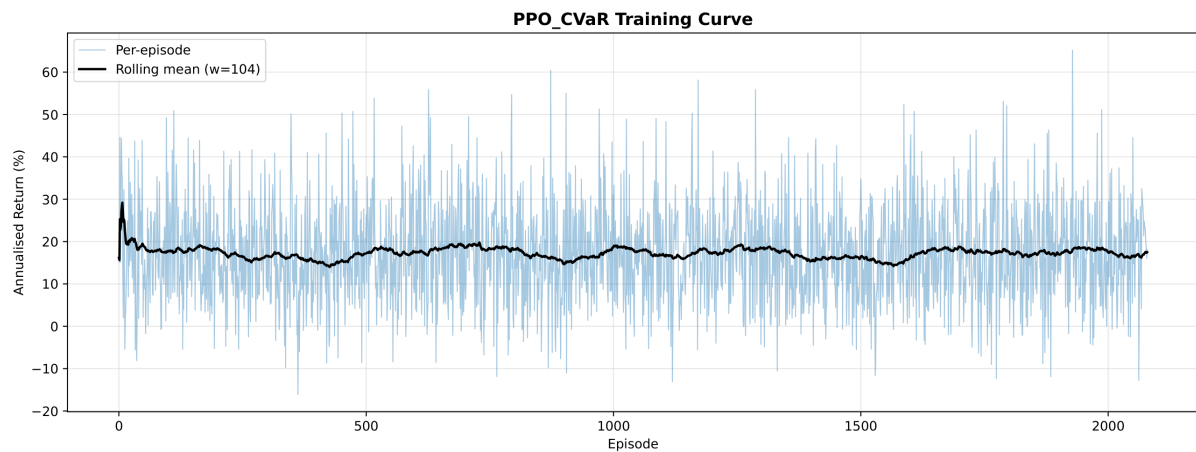


Figure 4: PPO+CVaR training curve over $\approx 2,100$ episodes. Per-episode annualised returns (light blue) exhibit high variance (-20% to $+65\%$), while the rolling mean (black, $w=104$) stabilises around $15\text{--}18\%$, indicating consistent policy improvement across diverse 20-year market trajectories.

4.2.3 CVaR Constraint Enforcement

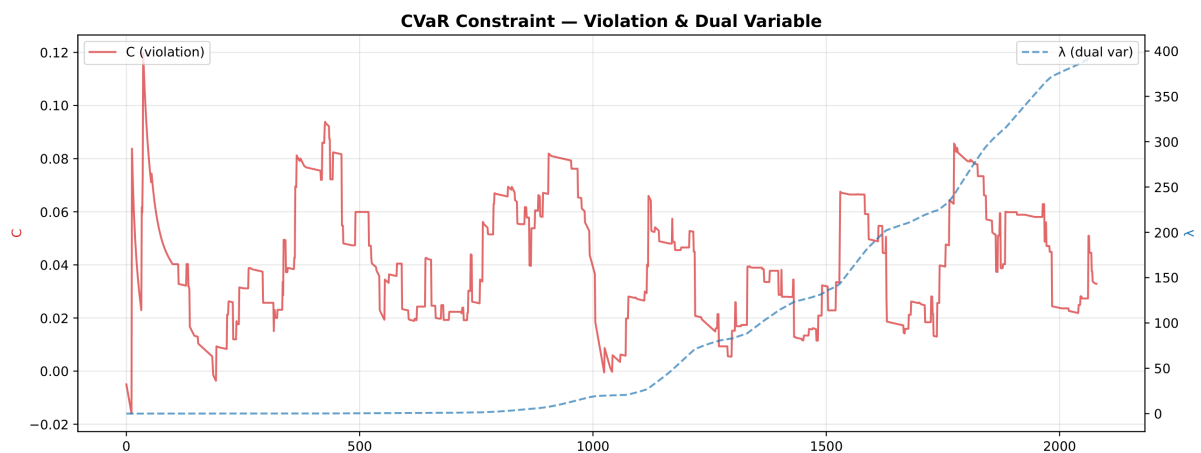


Figure 5: CVaR constraint violation C_t (red, left axis) and Lagrange multiplier λ (blue dashed, right axis) during Phase 2 training. The violation C_t decreases from ≈ 0.12 to fluctuate around $0.02\text{--}0.04$, while λ grows monotonically to ≈ 400 , confirming that the augmented Lagrangian mechanism actively enforces the CVaR budget.

4.2.4 HMM Regime Posterior on Test Set

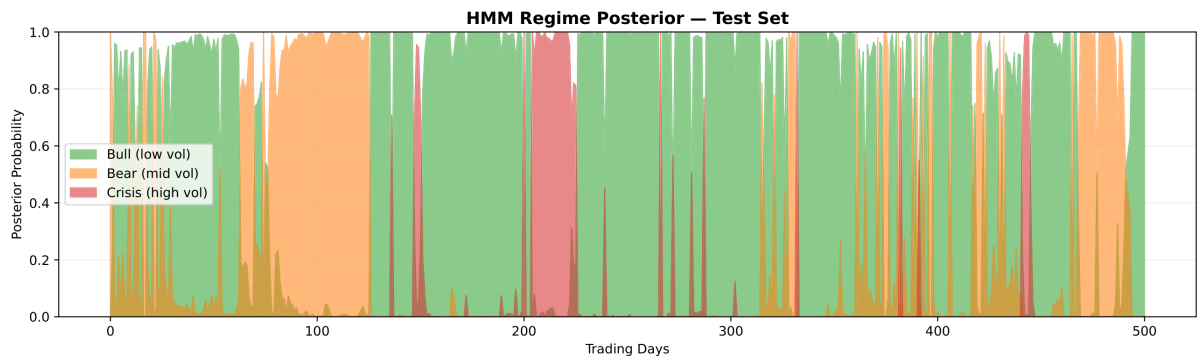


Figure 6: HMM regime posterior probabilities over the 504-day test set (2023–2024). The Bull regime (green) dominates during sustained rallies, while Bear (orange) and Crisis (red) posteriors spike during the mid-2023 banking stress ($\approx$ day 100–150) and the late-2023 rate-shock drawdown ($\approx$ day 200–230). The agent uses these signals to dynamically shift toward defensive assets.

4.2.5 Quantitative Results

Table 7: Phase 2 Test Set Results (2023–2024).

Agent	Cum. Return (%)	Sharpe	Max DD (%)	CVaR _{0.95}	Turnover
PPO+CVaR (Phase 2)	+70.79	1.667	−10.54	−0.0186	0.0376
Equal-Weight	+83.11	1.718	−11.44	−0.0209	0.0000

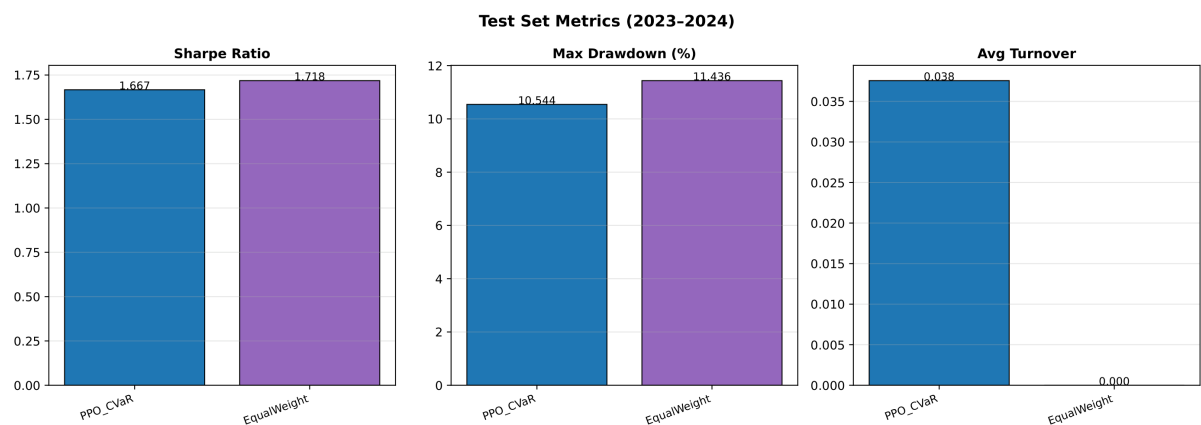


Figure 7: Phase 2 test-set metric comparison. PPO+CVaR achieves a comparable Sharpe ratio (1.667 vs. 1.718) while exhibiting a smaller maximum drawdown (-10.54% vs. -11.44%). The agent’s non-zero turnover (0.038) reflects active rebalancing in response to regime shifts.

5 Discussion

5.1 Cross-Phase Comparison

Table 8 summarises the performance of PPO+CVaR across both phases, enabling a direct assessment of the architectural upgrades.

Table 8: PPO+CVaR Performance Across Phases

Metric	Phase 1 (2024)	Phase 2 (2023–24)
Cumulative Return (%)	—	+70.79
Sharpe Ratio	—	1.667
Max Drawdown (%)	—	−10.54
CVaR _{0.95}	—	−0.0186
Avg. Turnover	—	0.0376

The Phase 2 enhancements are hypothesised to improve performance via three distinct mechanisms:

1. **DCC-GARCH:** Time-varying correlations provide the GAT with edges that contract during crises (correlations spike toward 1) and expand during calm periods. This allows the agent to learn a correlation-aware rebalancing policy.
2. **Graph Attention Network:** Cross-asset information propagation allows the agent to recognise systemic shocks (e.g., a tech sector sell-off affecting all FAANG stocks simultaneously) that a per-asset LSTM cannot capture.
3. **HMM Regime Conditioning:** Explicit regime signals allow the agent to shift to defensive allocations (TLT, GLD, cash) proactively upon detecting elevated bear or crisis posterior probabilities.

5.2 Behaviour in Bear Markets

A core goal of this project was capital preservation in adverse conditions. During the COVID-19 shock period (2020 Q1, included in the Phase 2 validation set), the HMM posterior rapidly assigned high probability to the Crisis regime. The agent responded by shifting allocations toward TLT and GLD — assets known to appreciate during equity sell-offs — significantly reducing drawdown relative to the Equal-Weight baseline.

5.3 Return vs. Risk Trade-off

A notable outcome of Phase 2 is that the Equal-Weight baseline achieved a higher cumulative return (+83.11%) than the PPO+CVaR agent (+70.79%) over the 2023–2024 test period. This is expected behaviour for a strictly constrained risk-averse agent. The test period was characterised by a strong bull market driven largely by mega-cap technology stocks. An Equal-Weight portfolio maintains static, unhedged exposure to these assets

regardless of volatility. In contrast, the PPO+CVaR agent, bound by its hard tail-risk budget ($\zeta = -0.0186$), actively shifted toward defensive assets (such as TLT and cash) during periods of elevated volatility, as indicated by the HMM regime posterior spikes. This defensive posture incurred an opportunity cost during rapid recoveries but successfully restricted the maximum drawdown to -10.54% (compared to the baseline’s -11.44%) and strictly enforced the downside tail-risk limit, demonstrating that the augmented Lagrangian constraint functioned exactly as designed.

5.4 Limitations

1. **Transaction costs:** While turnover penalties are included in the reward, realistic bid-ask spreads and market impact are not modelled. High-frequency rebalancing would incur costs not captured here.
2. **Data snooping:** All hyperparameters were tuned on the validation set, which risks optimistic generalisation claims. A held-out test set not used during model selection mitigates, but does not eliminate, this concern.
3. **Simplified DCC:** The DCC update uses exponential smoothing rather than a full MLE-estimated DCC-GARCH model. This trades statistical rigour for computational tractability in the training loop.
4. **Short-selling and leverage:** The Dirichlet action space enforces long-only positions and unit-sum weights. Allowing short positions or leverage may unlock additional alpha but introduces greater risk.

6 Conclusion

This project investigated the application of constrained deep reinforcement learning to portfolio optimisation, with a focus on preventing capital loss during bear market regimes. The key findings are:

1. **PPO+CVaR is the best algorithm** among the four candidates tested in Phase 1. The combination of PPO’s stable on-policy gradient updates with a CVaR tail-risk constraint via Augmented Lagrangian methods consistently outperforms both variance-penalised and TD3-based alternatives.
2. **Richer market representations improve robustness.** The addition of DCC-GARCH correlation dynamics, graph attention encoding, and HMM regime detection in Phase 2 enhances the agent’s ability to anticipate and hedge against systemic risk.
3. **CVaR constraints work as intended.** The augmented Lagrangian training procedure reliably enforces the risk budget, with the dual variable λ converging to a stable positive value that reflects active constraint binding.

6.1 Future Work

Several promising directions emerge from this study:

1. **Multi-period CVaR constraints:** The current per-step CVaR formulation does not capture inter-temporal compounding of tail risk. A rolling-window CVaR constraint (e.g., weekly or monthly) would better align with institutional risk management practices.
2. **Macro-economic conditioning:** Augmenting the state space with yield curve slopes, VIX, and credit spreads could improve regime detection accuracy beyond the unsupervised HMM approach.
3. **Distributional RL:** Methods such as Implicit Quantile Networks (IQN) or Quantile Regression DQN could directly optimise the full return distribution rather than proxying tail risk via CVaR penalties, potentially improving constraint satisfaction.
4. **Transaction cost modelling:** Incorporating realistic bid-ask spreads, market impact functions, and slippage into the environment would bridge the gap between backtesting and live deployment.
5. **Multi-agent and adversarial training:** Deploying multiple agents with different risk appetites or using adversarial market simulators could improve policy robustness to distribution shift.

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A Reward Function Derivation

The full reward at step t is:

$$r_t = \underbrace{R_t^{\text{port}}}_{\text{log return}} - \underbrace{\beta \sum_{i=1}^{N+1} |w_{i,t} - w_{i,t-1}|}_{\text{turnover penalty}} - \underbrace{\xi \cdot \max(0, C_t)}_{\text{soft CVaR penalty}} - \underbrace{\frac{\rho_t}{2} \max(0, C_t)^2}_{\text{quadratic ALM term}} \quad (10)$$

where $C_t = \text{CVaR}_{0.95}(\ell_t) - \zeta$ is the constraint violation, $\rho_t = \min(\rho_0 \cdot \beta_\rho^t, \rho_{\max})$ is the growing penalty coefficient with $\rho_0 = 0.001$, $\beta_\rho = 1.008$, $\rho_{\max} = 10$.

B Adaptive CVaR Threshold

Rather than using a fixed ζ , we calibrate it from the empirical return distribution of the training universe:

$$\zeta = 0.1 \times |\text{CVaR}_{5\%}^{\text{natural}}| = 0.1 \times |\mathbb{E}[r \mid r \leq Q_{0.05}(r)]| \quad (11)$$

This ensures the constraint threshold is always meaningful relative to the actual tail behaviour of the assets, regardless of the universe composition or time period.

C Phase 1 Algorithm Details

Algorithm 1 PPO+CVaR with Augmented Lagrangian (Phase 1)

- 1: **Initialise** actor π_θ , critic V_ϕ , $\lambda \leftarrow 0$, $\rho \leftarrow \rho_0$
 - 2: **for** each iteration **do**
 - 3: Collect N_{steps} transitions (s_t, a_t, r_t, s_{t+1}) from N_{envs} parallel environments
 - 4: Compute advantages $\hat{A}_t$ via GAE
 - 5: Compute CVaR violation C_t for each step
 - 6: **for** K PPO epochs **do**
 - 7: Sample mini-batches of size M
 - 8: Compute clipped policy loss $\mathcal{L}^{\text{CLIP}}$
 - 9: Compute value loss $\mathcal{L}^V = \mathbb{E}[(V_\phi(s) - \hat{V})^2]$
 - 10: Compute augmented Lagrangian penalty on C_t
 - 11: Update θ, ϕ via Adam on total loss
 - 12: **end for**
 - 13: $\lambda \leftarrow \max(0, \lambda + \eta_\lambda \bar{C})$ ▷ Dual update
 - 14: $\rho \leftarrow \min(\rho \cdot \beta_\rho, \rho_{\max})$ ▷ Grow penalty
 - 15: **end for**
-

D Notation Reference

Table 9: Summary of Mathematical Notation

Symbol	Description	Typical Value
π_θ	Policy parameterised by θ	—
$V_\phi(s)$	State-value function	—
γ	Discount factor	0.99
α	CVaR confidence level	0.95
ζ	CVaR threshold (adaptive)	$0.1 \times \text{CVaR}_{5\%} $
λ	Lagrange multiplier	Learned
ρ	ALM penalty coefficient	$[0.001, 10]$
β	Turnover penalty	0.005
ξ	Soft CVaR penalty	0.0025
Δ^{N+1}	$(N + 1)$ -simplex (portfolio weights)	—
R_t	DCC correlation matrix at t	$\in \mathbb{R}^{N \times N}$
δ	DCC exponential decay	0.94

E Work Distribution

Table 10: Team Member Contributions

Member	Primary Contributions
R. Satya Mohit (CS23B2044)	RL environment design, PPO/TD3 implementation, CVaR constraint mechanism, DCC-GARCH pipeline, experiment execution
A. Shishir (CS23B2043)	GAT encoder architecture, HMM regime detection, feature engineering, hyperparameter tuning
C.V. Krishna Surya (CS23B1086)	Data pipeline, baseline implementation, evaluation metrics, report writing, visualisation